

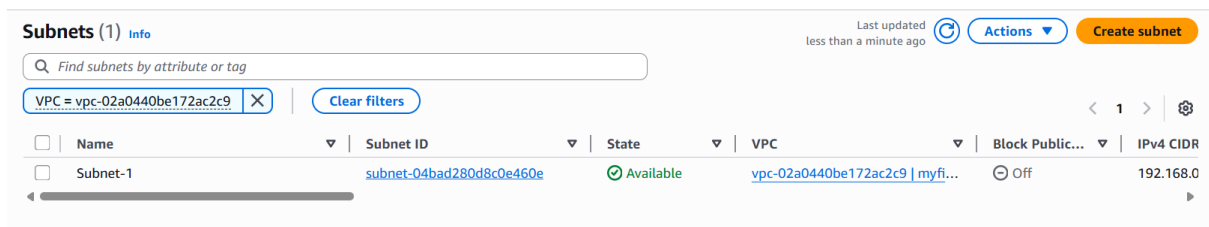
# Creating Multiple Subnets

## To Begin with the Lab

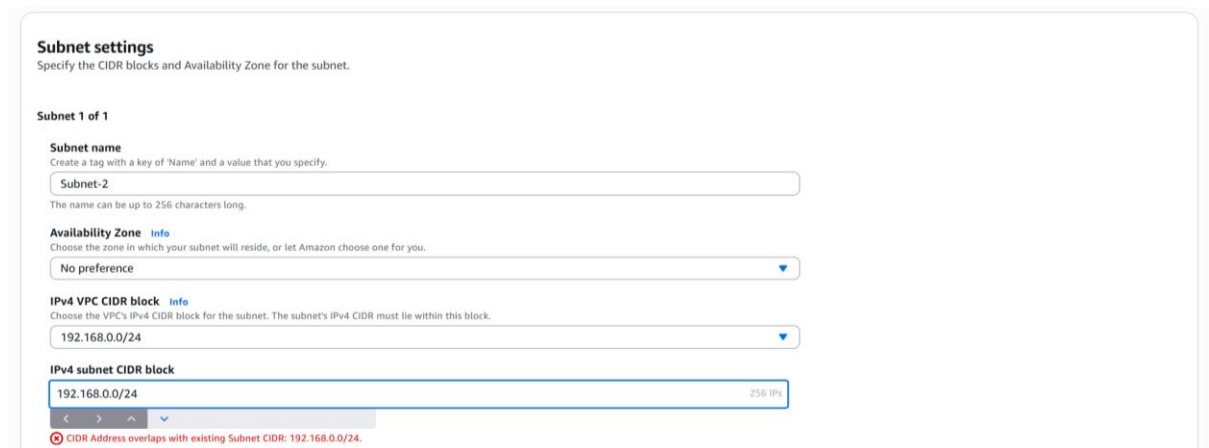
### Summary of the Lab

This lab demonstrated how to create multiple subnets within a custom VPC by dividing the VPC CIDR block into two /25 subnets across different Availability Zones. Two EC2 instances were launched in separate subnets, highlighting high availability. Initially, internet access failed because the VPC lacked an Internet Gateway. After creating and attaching an Internet Gateway and updating the main route table with a default route (0.0.0.0/0), the EC2 instances successfully gained internet connectivity.

- Open the **AWS Management Console** and navigate to the **VPC** service. Review the previously created VPC and subnet to understand that only one subnet currently exists, which limits the deployment of EC2 instances to a single Availability Zone.



- Attempt to create a second subnet in a different Availability Zone using the same CIDR block as the first subnet. AWS will display an error because the entire VPC CIDR range has already been assigned to the first subnet, leaving no available IP addresses for another subnet.



- Delete the previously created subnet that was using the entire /24 CIDR block so that the VPC address space becomes available again. This step is required because the original subnet occupies the complete IP range and prevents additional subnet creation.
- Create the first subnet by selecting the custom VPC, assigning it the name **Subnet-1**, choosing the **us-east-1a** Availability Zone, and specifying the CIDR block 192.168.0.0/25. This subnet will use the first half of the VPC IP address range.

Create subnet

Info

VPC

VPC ID

Create subnets in this VPC.

vpc-02a0440be172ac2c9 (myfirst-vpc)

Associated VPC CIDRs

IPv4 CIDRs

192.168.0.0/24

VPC

Subnets

Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Subnet-1

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1a)

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.0/25

128 IPs

< > ^ v

Tags - optional

Remove

- Create a second subnet in the same VPC by assigning it the name **Subnet-2**, selecting the **us-east-1b** Availability Zone, and specifying the CIDR block 192.168.0.128/25. This subnet will use the remaining half of the VPC IP address range.

VPC

Info

VPC

VPC ID

Create subnets in this VPC.

vpc-02a0440be172ac2c9 (myfirst-vpc)

Associated VPC CIDRs

IPv4 CIDRs

192.168.0.0/24

VPC

Subnets

Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Subnet-2

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az6 (us-east-1b)

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/24

IPv4 subnet CIDR block

192.168.0.128/25

128 IPs

< > ^ v

Tags - optional

Remove

- Verify that both subnets have been created successfully and are located in different Availability Zones. This configuration allows workloads to be distributed across multiple Availability Zones for higher availability and fault tolerance.

**Subnets (2)** [Info](#) Last updated 3 minutes ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

VPC = vpc-02a0440be172ac2c9 [Clear filters](#)

| <input type="checkbox"/> | Name     | Subn...   | State     | VPC        | Bloc... | IPv4 CIDR        | IPv6 ... | IPv6 ... | Avail... | Availability Zone     |
|--------------------------|----------|-----------|-----------|------------|---------|------------------|----------|----------|----------|-----------------------|
| <input type="checkbox"/> | Subnet-2 | subnet... | Available | vpc-02a... | Off     | 192.168.0.128/25 | -        | -        | 123      | use1-az6 (us-east-1b) |
| <input type="checkbox"/> | Subnet-1 | subnet... | Available | vpc-02a... | Off     | 192.168.0.0/25   | -        | -        | 123      | use1-az4 (us-east-1a) |

- Open the EC2 console and launch the first EC2 instance.

**EC2** > **Instances** Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

| <input type="checkbox"/> | Name | Instance ID | Instance state | Instance type | Status check | Availability Zone |
|--------------------------|------|-------------|----------------|---------------|--------------|-------------------|
|                          |      |             |                |               |              |                   |

- Name the Instance and choose the AMI

**Launch an instance** [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

**Name and tags** [Info](#)

Name:  [Add additional tags](#)

**Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)  
Including AMIs from AWS, Marketplace and the Community

**Summary**

[Cancel](#) [Launch instance](#) [Preview code](#)

- Choose the Instance type and create a new key-pair if it is available for you

**Instance type** [Info](#) [Get advice](#)

Instance type: **t3.micro** Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

**Summary**

[Cancel](#) [Launch instance](#) [Preview code](#)

**Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*:  [Create new key pair](#)

- Select the custom VPC, choose **Subnet-1**, enable **Auto-assign Public IP** if required, and complete the instance launch process.

EC2 > Instances > Launch an instance

**Network settings** Info

VPC - required Info

vpc-02a0440be172ac2c9 (myfirst-vpc) 192.168.0.0/24

Subnet Info

subnet-00a2151c33083cd00 Subnet-1 Create new subnet

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Security group name - required

launch-wizard-31

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and \_-./!@,[]+=&()|!\*~

Description - required Info

launch-wizard-31 created 2026-06-29T10:55:26.944Z

Inbound Security Group Rules

**Summary**

Cancel Launch instance Preview code

- Now click Launch instance

Instances (1) Info

Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

VPC ID = vpc-02a0440be172ac2c9 Clear filters

|                          | Name     | Instance ID         | Instance state | Instance type | Status check | Availability Zone | Public IP |
|--------------------------|----------|---------------------|----------------|---------------|--------------|-------------------|-----------|
| <input type="checkbox"/> | Server-1 | i-0ff79ae0e6c28815a | Running        | t3.micro      | Initializing | us-east-1a        | -         |

- Before launching the second EC2 instance, open the **Subnet Settings** for **Subnet-2** and enable **Auto-assign Public IPv4 Address**. This ensures that any EC2 instance launched in this subnet automatically receives a public IP address.

subnet-027492ac7e08ce09b / Subnet-2

**Details**

Subnet ID  
subnet-027492ac7e08ce09b

IPv4 CIDR  
192.168.0.128/25

Subnet ARN  
arn:aws:ec2:us-east-1:463646775279:subnet/subnet-027492ac7e08ce09b

Available IPv4 addresses  
192.168.0.128/25

State  
Available

IPv6 CIDR  
-

Actions

Create flow log

Edit subnet settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Edit CIDR association

VPC > Subnets > subnet-027492ac7e08ce09b > Edit subnet settings

**Edit subnet settings** Info

Subnet

Subnet ID  
subnet-027492ac7e08ce09b

Name  
Subnet-2

**Auto-assign IP settings** Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address Info

☐ Enable auto-assign customer-owned IPv4 address Info

Option disabled because no customer owned pools found.

- In the same way create the second EC2 Instance
- Launch a second EC2 instance by selecting the custom VPC and choosing **Subnet-2**. Since public IP assignment is enabled for this subnet, the instance will automatically receive a public IP during launch.

EC2 > Instances > Launch an instance

### Network settings

**VPC** - required [Info](#)

vpc-02a0440be172ac2c9 (myfirst-vpc) [192.168.0.0/24](#)

**Subnet** [Info](#)

subnet-027492ac7e08ce09b Subnet-2 [Create new subnet](#)

VPC: vpc-02a0440be172ac2c9 Owner: 463646775279  
Availability Zone: us-east-1b (use1-az6) Zone type: Availability Zone  
IP addresses available: 125 CIDR: 192.168.0.128/25

**Auto-assign public IP** [Info](#)

Enable

Additional charges apply when outside of free tier allowance

**Firewall (security groups)** [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

launch-wizard-32

### Summary

**Number of instances** [Info](#)

1

**Software Image (AMI)**

Amazon Linux 2023 AMI 2023.12...[read more](#)  
ami-08f44e8eca9095668

**Virtual server type (instance type)**

t3.micro

**Firewall (security group)**

New security group

**Storage (volumes)**

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

- Two instances are running

Instances (2) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#)

VPC ID = vpc-02a0440be172ac2c9 [Clear filters](#)

| <input type="checkbox"/> | Name <a href="#">Info</a> | Instance ID         | Instance state                                       | Instance type | Status check      | Availability Zone | Public IP |
|--------------------------|---------------------------|---------------------|------------------------------------------------------|---------------|-------------------|-------------------|-----------|
| <input type="checkbox"/> | Server-2                  | i-0fb1e245e2b5e3b04 | Running <a href="#">Info</a> <a href="#">Actions</a> | t3.micro      | 3/3 checks passed | us-east-1b        | -         |
| <input type="checkbox"/> | Server-1                  | i-0ff79ae0e6c28815a | Running <a href="#">Info</a> <a href="#">Actions</a> | t3.micro      | 3/3 checks passed | us-east-1a        | -         |

- You can see the IP address attached to Server-1 Instance
- Attempt to connect to the EC2 instances using SSH with their public IP addresses. Although the instances have public IPs, the connection will fail because the custom VPC does not yet have an **Internet Gateway** attached.

```
ubuntu@LAPTOP-70U805T1:~$ ssh -i jenkins.pem ec2-user@54.234.0.98
ssh: connect to host 54.234.0.98 port 22: Connection timed out
ubuntu@LAPTOP-70U805T1:~$
```

- Also, if you try to connect it using AWS console, still it will fail

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

**Instance is not in public subnet**

Associated subnet [subnet-00a2151c33083cd00 \(Subnet-1\)](#) is not a public subnet.  
To use EC2 Instance Connect, your instance must be in a public subnet. [To make the subnet a public subnet, add a route in the subnet route table to an internet gateway.](#)

Instance ID

[i-0ff79ae0e6c28815a \(Server-1\)](#)

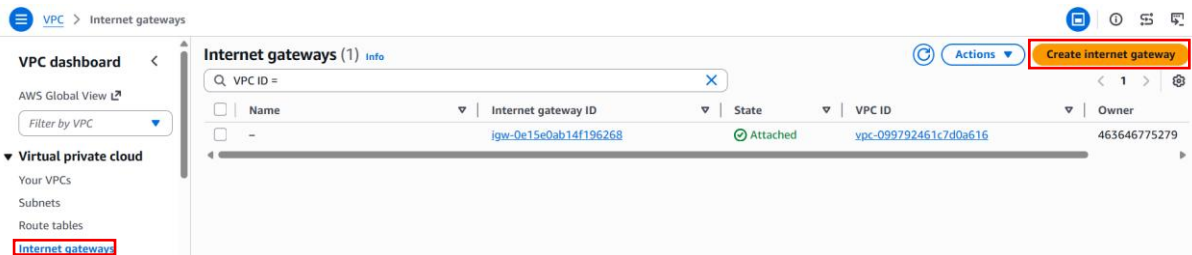
**Connection type**

☒ Connect using a Public IP  
Connect using a public IPv4 or IPv6 address

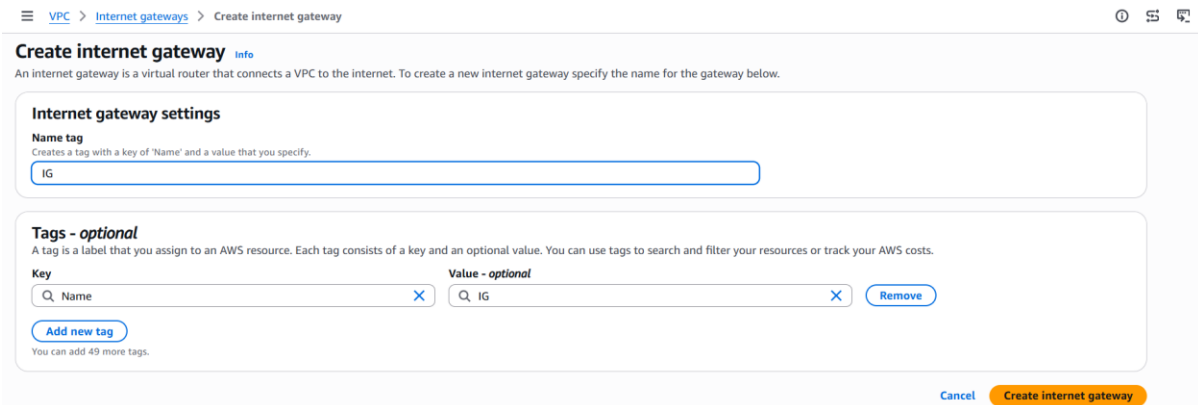
☐ Connect using a Private IP  
Connect using a private IP address and a VPC endpoint

**Failed to connect to your instance**  
Error establishing SSH connection to your instance. Try again later.

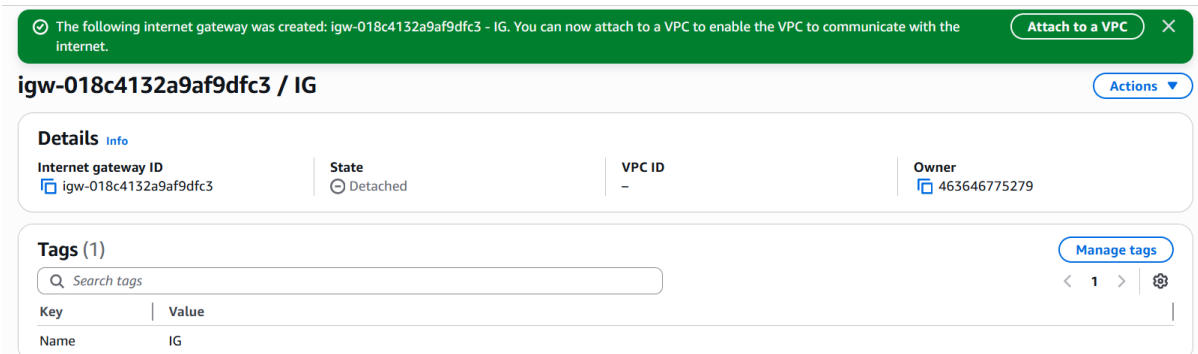
- Return to the **VPC** console and select **Internet Gateways** from the navigation pane. Click **Create Internet Gateway**, provide a meaningful name such as **IGW**, and create the Internet Gateway.



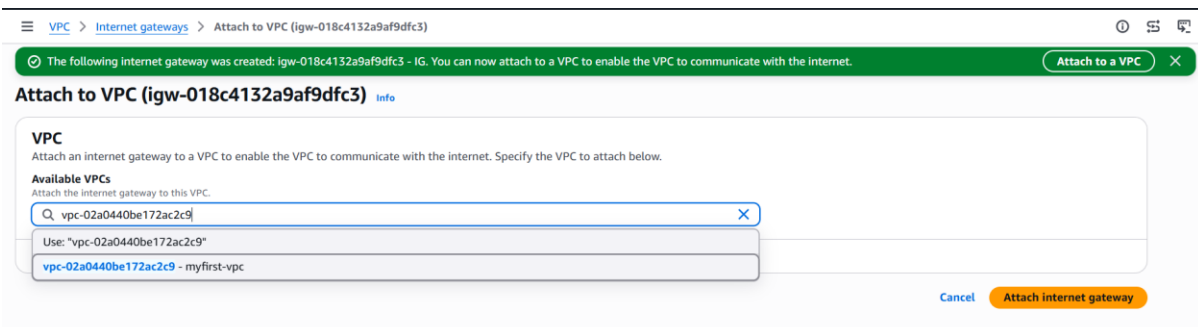
- Give the name and click create internet gateway.



- After creating the Internet Gateway, select it and choose **Actions** → **Attach to VPC**. Select your custom VPC and attach the Internet Gateway so that the VPC can communicate with the internet.

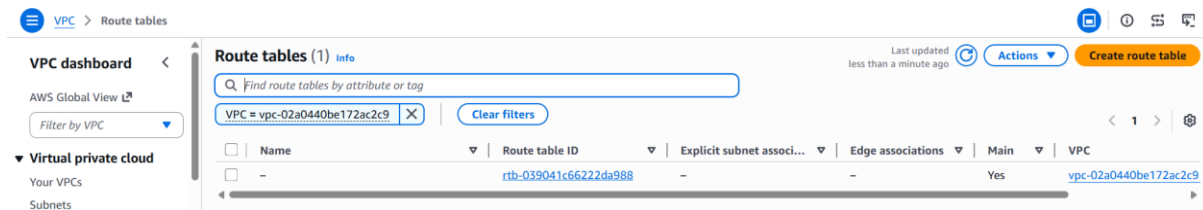


- Or you can directly click Attach to a VPC.

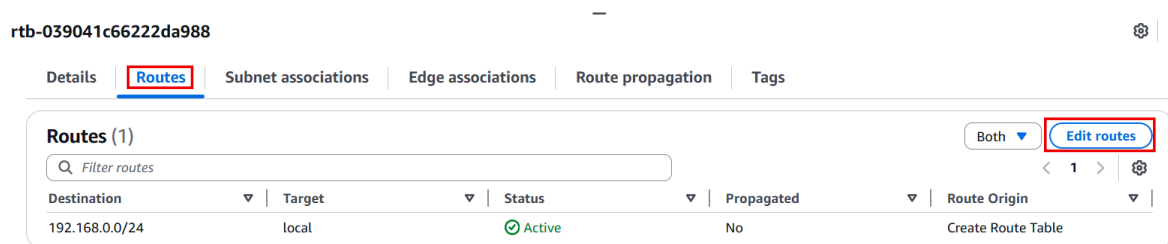


- Now go to the VPC Dashboard
- Click on the Route table

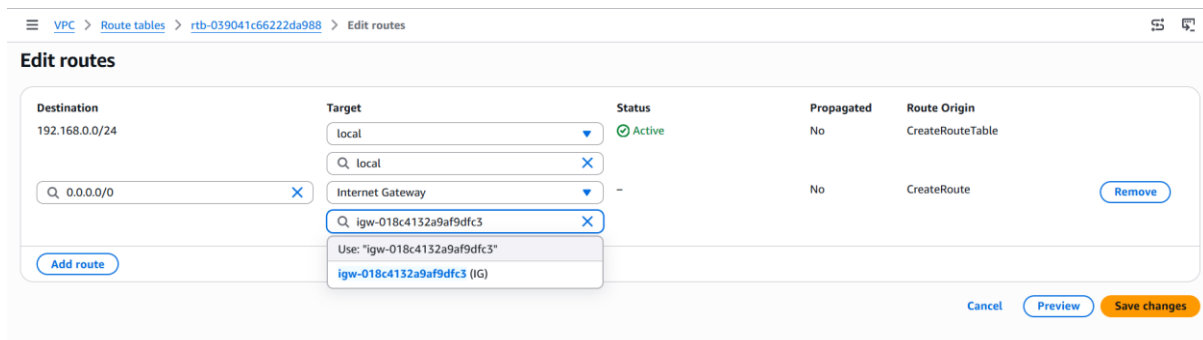
- Open the **Route Tables** section in the VPC console and locate the **Main Route Table** that is automatically associated with your VPC. This route table is created by AWS and cannot be deleted, and all subnets are associated with it by default.



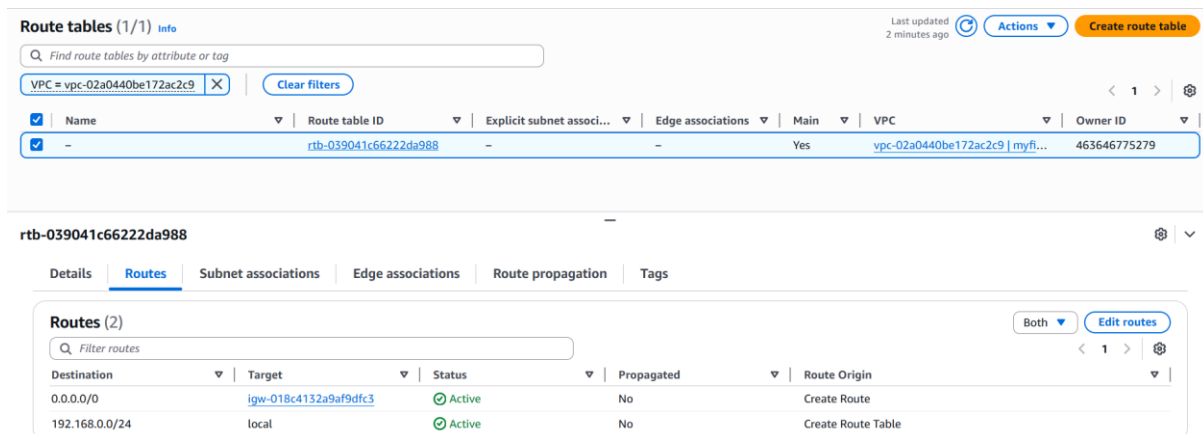
- Select the Main Route Table and open the **Routes** tab, then click **Edit Routes**.



- Add a new route with the destination **0.0.0.0/0** and set the target as the Internet Gateway you created, then save the changes.



- Verify that the newly added route shows an **Active** status. This confirms that all internet-bound traffic from the associated subnets will now be routed through the Internet Gateway.



Let's retry to connect again

```
ubuntu@LAPTOP-70U805T1:-$ ssh -i jenkins.pem ec2-user@54.234.0.98
```



```
##### Amazon Linux 2023
#####
#####
#/
V/m ^ i->
https://aws.amazon.com/linux/amazon-linux-2023
m/
[ec2-user@ip-192-168-0-88 ~]$ |
```

You can see the we can access the google.com

```
ubuntu@LAPTOP-70U80ST1:~$ ssh -i jenkins.pem ec2-user@54.234.0.98
ssh: connect to host 54.234.0.98 port 22: Connection timed out
ubuntu@LAPTOP-70U80ST1:~$ ssh -i jenkins.pem ec2-user@54.234.0.98
```



Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
Last login: Mon Jun 29 11:48:40 2026 from 202.83.29.50
[ec2-user@ip-192-168-0-88 ~]$ ping google.com
PING google.com (142.251.179.101) 56(84) bytes of data.
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=1 ttl=186 time=1.69 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=2 ttl=186 time=1.74 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=3 ttl=186 time=1.71 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=4 ttl=186 time=1.74 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=5 ttl=186 time=1.60 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=6 ttl=186 time=1.55 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=7 ttl=186 time=1.57 ms
 64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=8 ttl=186 time=1.72 ms
^C
--- google.com ping statistics ---
 8 packets transmitted, 8 received, 0% packet loss, time 7010ms
rtt min/avg/max/mdev = 1.552/1.663/1.736/0.071 ms
[ec2-user@ip-192-168-0-88 ~]$
```